

- RAG Learning

P5 — Radioactive Materials

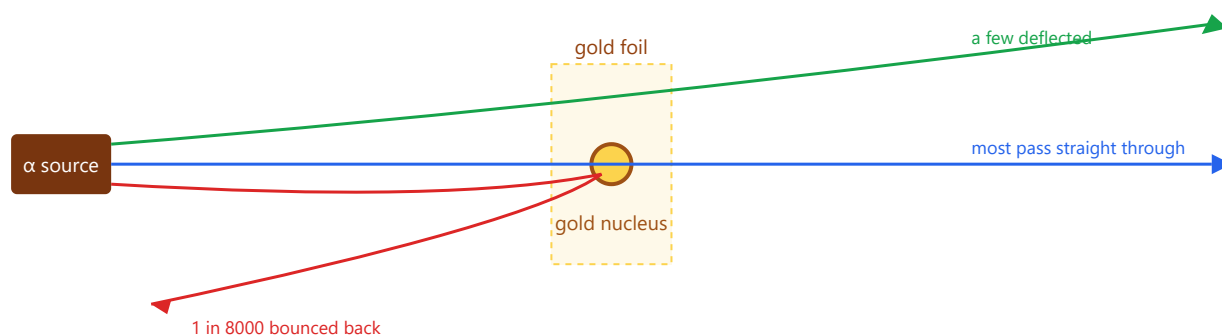
OCR GCSE Physics B (J259) — 21st Century Science · Paper 2 (Depth in Physics)

P5.1 The Atomic Model — A Historical Story

Ideas about the atom have changed over time as experimental evidence accumulated. You should know the sequence of models:

Model	Scientist(s)	Key idea
Tiny indivisible particles ("atomos")	Democritus ($\approx$ 400 BC)	Matter made of indivisible units
Sphere of positive matter with negative electrons dotted through it	J.J. Thomson, 1897 — plum pudding model	Discovery of the electron
Small, dense, positive nucleus with electrons around it	Rutherford, 1911 — gold foil / α-scattering experiment	Most of the atom is empty space; nearly all its mass is in a tiny central nucleus
Electrons orbit in fixed shells (energy levels)	Bohr, 1913	Explains line spectra
Nucleus contains protons and neutrons	Chadwick, 1932	Discovery of the neutron

Rutherford's α -particle Scattering Experiment

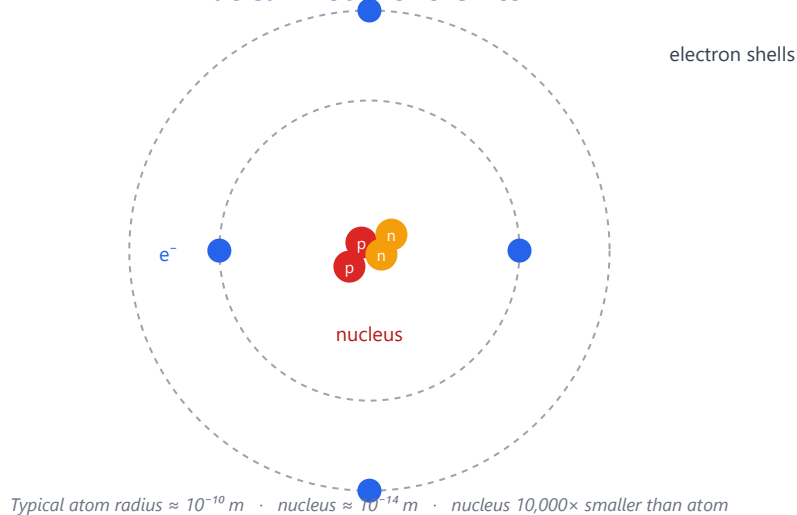


Exam Tip: Rutherford's **conclusions** from the results:

- Most α pass straight through $\rightarrow$ atoms are mostly empty space.
- A few deflected at large angles $\rightarrow$ small, dense, *positively charged* nucleus.
- 1 in 8000 bounced back $\rightarrow$ nucleus is very small but contains nearly all the mass.

P5.2 Structure of the Atom

Nuclear Model of the Atom



Particle	Relative mass	Relative charge	Location
Proton (p)	1	+1	Nucleus

Neutron (n)	1	0	Nucleus
Electron (e ⁻)	1/1836 (negligible)	-1	Shells / orbitals

Nuclear Notation

An atom is written as ${}^A_Z\text{X}$ where:

- **Z** = proton (atomic) number — number of protons; defines the element.
- **A** = mass (nucleon) number — total protons + neutrons.
- Number of neutrons = $A - Z$.

Isotopes are atoms of the same element with the same Z but different A (different number of neutrons). Example: hydrogen has three isotopes — ${}^1\text{H}$, ${}^2\text{H}$ (deuterium), ${}^3\text{H}$ (tritium).

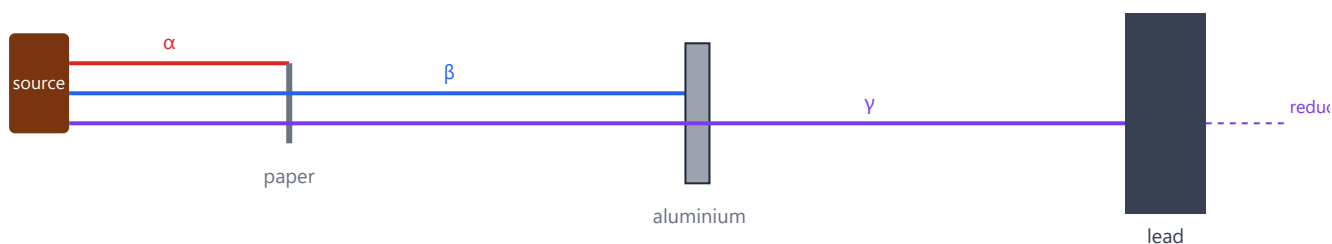
P5.3 Types of Nuclear Radiation

Some nuclei are **unstable** — they decay spontaneously, emitting radiation. The three main types you need are **alpha (α)**, **beta (β)** and **gamma (γ)**. Positrons (β^+) and free neutrons are also mentioned at GCSE.

Radiation	What it is	Charge	Mass	Stopped by	Ionising
Alpha (α)	Helium nucleus: ${}^4_2\text{He} \rightarrow 2\text{p} + 2\text{n}$	+2	4 u	A few cm of air; paper; skin	Very strongly ionising
Beta-minus (β^-)	Fast electron from the nucleus (a neutron $\rightarrow$ proton + e ⁻)	-1	≈ 0	A few mm of aluminium	Moderately ionising
Gamma (γ)	High-frequency EM wave	0	0	Several cm of lead / metres of concrete (reduced, never fully)	Weakly ionising

Positron (β^+)	Antiparticle of the electron (proton $\rightarrow$ neutron + β^+)	+1	≈ 0	Annihilates with an electron almost immediately	—
Neutron	Free neutron (some decays)	0	1	Water / hydrogen-rich materials	Via induced radioactivity

Penetration of α , β and γ



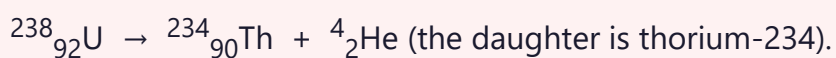
P5.4 Nuclear Equations

α decay: ${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\text{He}$
(mass falls by 4, atomic number falls by 2)

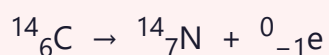
β^- decay: ${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0\text{e}$
(mass unchanged, atomic number rises by 1 — a neutron becomes a proton)

γ emission: no change in A or Z — the nucleus loses energy as a γ photon.

Worked Example: Uranium-238 decays by alpha emission. Write the nuclear equation.



Worked Example: Carbon-14 decays by β^- emission. Write the nuclear equation.

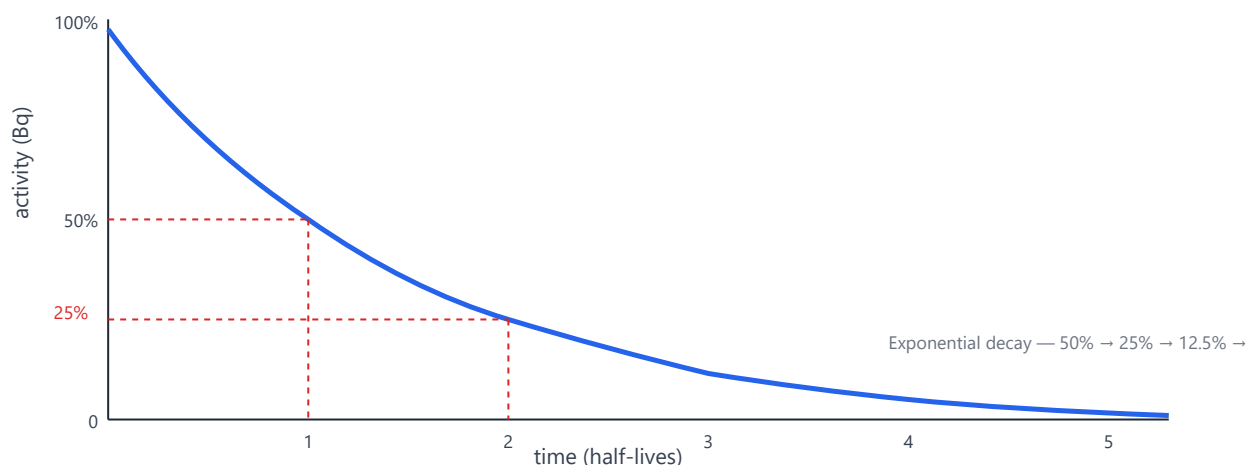


P5.5 Activity and Half-Life

Activity (A) = number of decays per second. Unit: **becquerel (Bq)**, 1 Bq = 1 decay/s.

Half-life ($t_{1/2}$) = the average time for the number of unstable nuclei in a sample — or equivalently the activity — to halve.

Radioactive Decay Curve



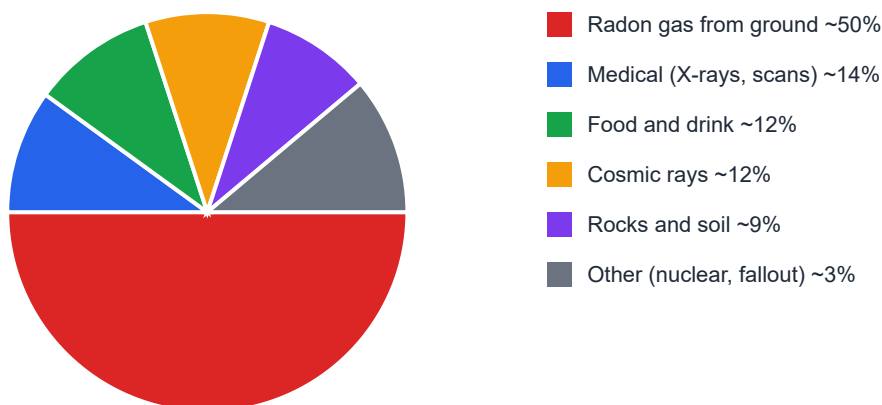
Worked Example: A sample of iodine-131 has an initial activity of 8000 Bq. Its half-life is 8 days. Find the activity after 24 days.

$24 \div 8 = 3$ half-lives. Activity = $8000 \times (1/2)^3 = 8000 / 8 = \mathbf{1000 \text{ Bq}}$.

P5.6 Background Radiation

There is always a low level of ionising radiation around us. Typical UK sources (approximate contributions):

Sources of UK Background Radiation



Background radiation must be **subtracted** from measured count rates when taking readings from a source, otherwise results are too high.

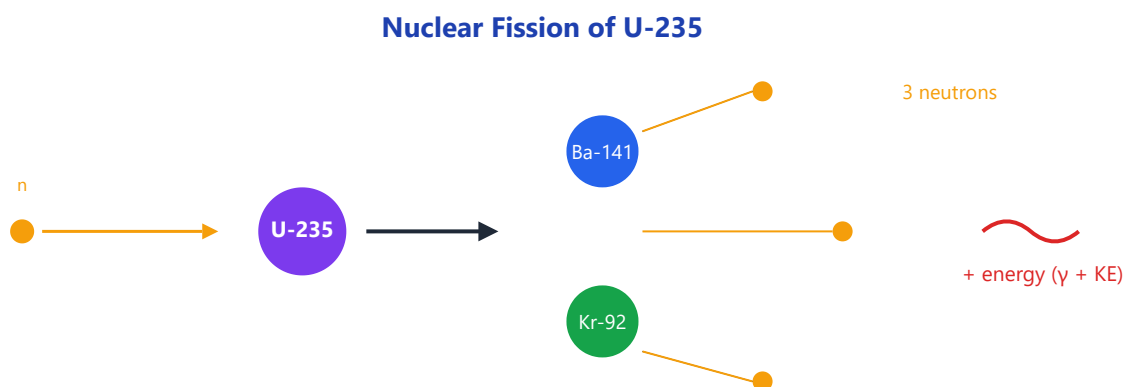
P5.7 Uses of Radiation in Medicine & Industry

Use	Radiation type	Why that type
Medical tracer (e.g. technetium-99m)	γ + short half-life (6 h)	Leaves body quickly; penetrates tissue to the detector
Radiotherapy for cancer	γ or high-energy X-ray (external); β (internal)	Kills cancer cells; directed from many angles to minimise healthy tissue dose
Sterilising surgical equipment / food	γ	Penetrates packaging; kills bacteria
Smoke detectors	α (americium-241)	Short range — safe in the home; smoke absorbs α , breaking a small ionisation current
Thickness gauge for paper / foil	β	Partially absorbed — change in transmitted count indicates thickness

Dating rocks / archaeology	U-238, C-14	Long, known half-lives
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P5.8 Nuclear Fission and Fusion

Fission: a large unstable nucleus (e.g. U-235) splits into two smaller, more stable nuclei, releasing a few neutrons and energy. The neutrons can trigger further fissions — a **chain reaction**.



Fusion: two small nuclei (e.g. hydrogen isotopes) collide at very high speed and join to make a larger nucleus (helium), releasing energy. This is how the Sun and all stars shine.

Fusion releases *much more energy per kg* than fission but requires extremely high temperatures and pressures to overcome electrostatic repulsion between the nuclei. Sustainable Earth-based fusion power is still being developed (ITER, tokamak designs).

P5.9 Dangers of Ionising Radiation

- **Irradiation:** being exposed to radiation *from outside*. The object does not become radioactive.
- **Contamination:** radioactive material gets *onto or into* the body (skin, food, dust). Continues to irradiate from inside — much harder to stop.

Risk depends on the activity, the type of radiation, and the duration and location of exposure. Inside the body, **alpha** is the most dangerous because it is very strongly ionising; outside the body, **gamma** is the most dangerous because it penetrates the most.